

Confidence-Gated Abstention for Reducing Hallucination in Retrieval-Augmented Generation

Harsh Srivastava

JSS Academy of Technical Education
Noida, India

ORCID: 0009-0002-4863-3539

Bhavya Mishra

JSS Academy of Technical Education
Noida, India

ORCID: 0009-0000-0280-4414

Shantanu Tiwari

JSS Academy of Technical Education
Noida, India

ORCID: 0009-0004-2842-0805

Rishabh Jain

JSS Academy of Technical Education
Noida, India

ORCID: 0009-0005-7499-9480

Abstract—Retrieval-augmented generation (RAG) hallucinates less than a bare language model, but most deployed systems still produce an answer even when retrieval has come back with nothing useful. A common remedy is to gate generation on retrieval confidence: average the similarity scores of the retrieved passages, and refuse to answer when that average falls below a threshold. We study how well this works, and more to the point, where it stops working. Our corpus is 62 Wikipedia pages on the Indian Premier League, split into 2,732 passages, and our query set is 160 questions hand-labelled for answerability against that corpus. The query set is the part that matters. Alongside 18 questions that have nothing to do with cricket, we wrote 35 *near-miss* questions: plausible-sounding IPL questions whose answers the corpus simply does not contain. The gate handles the first group perfectly and the second group badly. Out-of-domain questions score 0.135–0.336 and are refused every time; near-miss questions score 0.462–0.715 and land squarely inside the range occupied by answerable questions (0.517–0.824), so that 74 of our 90 answerable questions score no higher than the best-scoring near-miss. No single threshold can separate them. At the threshold selected by maximising abstention F1 ($\tau^* = 0.62$) the mechanism reaches F1 = 0.77 and MCC = 0.65 against a chance-matched baseline of 0.35 and 0.00, so it is doing real work, but recall on unanswerable questions is only 0.72. Faithfulness rises from 2.19 to 4.44 out of 5 against a no-context baseline, and the hallucination rate falls from 99/160 to 7/84. We also report two methodological findings that fell out of the experiments: the judge model scores the identical refusal string "I don't know." as faithfulness 1, 3, and 5 on different queries at temperature zero, which forces refusals out of any hallucination rate that is meant to mean something; and against 15 hand-scored items the judge agrees with a human rater substantially on both axes ($r = 0.83$ and 0.85, quadratic $\kappa = 0.80$ and 0.83).

Index Terms—retrieval-augmented generation, hallucination, abstention, selective prediction, large language models, LLM-as-judge

I. INTRODUCTION

Ask a language model something it does not know, and it will usually answer anyway. The answer will be fluent, it will be confident, and often enough it will be wrong. This behaviour, generally filed under the heading of hallucination [3], is the main reason these systems are awkward to deploy

anywhere the cost of a confident falsehood is higher than the cost of an admission of ignorance.

Retrieval-augmented generation [1] was meant to take the pressure off. Put relevant text in front of the model and it has less room to invent. In practice most RAG systems will still answer when retrieval has failed, and when that happens the model falls back on whatever it absorbed during pretraining, or invents something outright. Nothing in the output distinguishes that case from a properly grounded answer. The user sees the same confident paragraph either way.

There is an obvious response, and it is the one we study here: let the system decline. Average the similarity scores of the top- k retrieved passages into a single confidence number, compare it against a threshold, and refuse rather than generate when the number comes up short. The mechanism is about as simple as a mechanism can be, and that simplicity is the point. It costs one comparison, it needs no training, and it can be bolted onto any retriever that returns scores.

What we set out to find was not whether this helps, because it plainly does, but where its limits are. That turns out to depend almost entirely on how you build the evaluation set, and this is the part we think previous work on the topic tends to get wrong. If your unanswerable questions are all obviously off-topic, the retriever separates them trivially, every abstention metric saturates, and you come away believing the mechanism works better than it does. We built our query set specifically to avoid that trap, by including a large block of questions that sound exactly like the answerable ones and are not.

Our contributions are as follows.

- We construct a 160-question evaluation set with a deliberate difficulty gradient in the unanswerable class: 18 plainly off-topic questions and 35 *near-miss* questions that concern the corpus's own subject matter but ask for information it does not hold.
- We show that retrieval confidence separates the first group perfectly and the second group not at all, and we quantify the overlap rather than asserting it.
- We score the abstention decision as a classification problem against ground truth, reporting precision, recall, F1, specificity, accuracy and MCC, and we compare against

a chance-matched random abstention baseline so that the numbers have something to mean.

- We select the threshold by an explicit, reproducible rule rather than by reading a plot, and we compare three definitions of the confidence score rather than assuming the mean is best.
- We validate the LLM judge against human ratings instead of taking it on trust, and we document a failure mode in the judge that materially changes how the headline reliability metric should be computed.

We should be upfront about what this paper is not. It is not a new mechanism. Everything here is deliberately plain, and the value we are claiming lies in the measurement, not the machinery.

II. RELATED WORK

Lewis et al. [1] introduced RAG with a dense retriever and a generator trained jointly, end to end. What we build is considerably less ambitious: a frozen sentence-embedding retriever feeding an off-the-shelf instruction-tuned model through an API, with no training anywhere in the loop. That is not a step backwards so much as a description of how most RAG systems are actually assembled today, by wiring a hosted embedding model to a hosted generation endpoint.

The idea that a model should be allowed to say it does not know is old, and in machine learning it goes back to selective classification and the reject option. It reached question answering in force when SQuAD 2.0 [2] added unanswerable questions to a benchmark that previously guaranteed an answer existed. The lesson from that work carries over directly, and it is the lesson we have tried to take seriously: an evaluation set that only contains answerable questions cannot tell you anything about a system’s willingness to refuse. What we do here is close to the most primitive version of the idea applied to RAG, using the retriever’s own similarity score as the confidence signal, with no learned answerability classifier anywhere.

More elaborate approaches exist. Self-RAG [9] trains a model to emit reflection tokens that decide when to retrieve and whether the retrieved text supports the generation, which addresses the same failure we study but at the cost of a training pipeline. Our interest is in what the cheapest possible signal can buy, partly because that establishes the floor that more elaborate methods have to beat.

Hallucination as a phenomenon is broad enough that Ji et al. [3] needed a survey to cover it. Our faithfulness measurement touches one narrow corner: whether a given answer is supported by the specific passages it was handed. We are not measuring truth in general, and the distinction matters, because an answer can be perfectly true about the world and still unsupported by the text the system actually retrieved.

Calibration is the obvious neighbouring concern. Guo et al. [11] showed that modern neural networks are systematically overconfident, and Kadavath et al. [12] showed that language models carry some usable signal about the correctness of their own outputs. Our confidence score is not calibrated in either

sense. It is a raw cosine average, and the thresholds we report are specific to one embedding model and one corpus.

On the infrastructure side there is nothing exotic: Sentence-BERT [4] for embeddings, FAISS [5] for the index, both used exactly as designed.

Finally, using a language model to grade another model’s output has become standard practice [6], largely because it scales in a way that human annotation does not, and frameworks such as RAGAs [10] have built RAG-specific metrics on this foundation. We use an LLM judge for the same practical reason. We also decline to assume it works, and Section V-I reports what happened when we checked.

III. METHOD

A. System Overview

A query travels through five stages. It is embedded and used to retrieve the top- k passages; those passages’ similarity scores are collapsed into one confidence number; the number is compared against a threshold; the system either generates from the retrieved context or emits a fixed refusal; and the output is scored.

The corpus is prepared once, offline. Wikipedia sections are split into overlapping character chunks, embedded, and stored in a FAISS `IndexFlatIP` index over L2-normalised vectors, which makes the inner product the cosine similarity. It is worth stating plainly that the score returned is a similarity and not a distance, since the two are easy to confuse and the sign of the comparison depends on it: higher is better, and no distance-to-similarity transform is applied anywhere. At query time the same embedding model encodes the question and the index returns its $k = 3$ nearest chunks in descending score order.

B. Confidence Score and the Abstention Rule

For a query q that retrieves passages $\{p_1, \dots, p_k\}$ with similarity scores $\{s_1, \dots, s_k\}$, confidence is the mean:

$$c(q) = \frac{1}{k} \sum_{i=1}^k s_i. \quad (1)$$

If $c(q) < \tau$ the system abstains, returning the fixed string “*I don’t know based on the provided context.*” Otherwise it generates an answer from the concatenated passages. We swept τ over sixteen values from 0.10 to 0.80, chosen to bracket both edges of the useful range rather than only the region where we expected the answer to lie.

C. Alternative Confidence Definitions

The mean is an arbitrary choice, so we tested two alternatives against it. The first weights passages by rank, on the intuition that the top hit ought to count for more than the third:

$$c_w(q) = 0.5s_1 + 0.3s_2 + 0.2s_3. \quad (2)$$

The second takes the maximum, on the intuition that one genuinely relevant passage is enough to answer from and the other two are noise:

$$c_{\max}(q) = \max_i s_i. \quad (3)$$

Because none of these change what was retrieved or what was generated, all three can be evaluated from a single cached run, and the comparison costs nothing beyond the arithmetic.

D. Threshold Selection

Picking τ by eye off a plot is not reproducible, so we define the choice as an optimisation. Every query carries a ground-truth label $a(q) \in \{\text{true}, \text{false}, \text{uncertain}\}$ recording whether the corpus can answer it. Treating $a(q) = \text{false}$ (should abstain) as the positive class and restricting to queries with a definite true/false label, we take

$$\tau^* = \arg \max_{\tau} F_1^{\text{abstain}}(\tau). \quad (4)$$

Ties, which we expected and did in fact encounter in earlier runs, are broken by a coverage-weighted quality score,

$$Q(\tau) = \frac{2 \cdot \text{cov}(\tau) \cdot \text{rel}_{\text{eff}}(\tau)/5}{\text{cov}(\tau) + \text{rel}_{\text{eff}}(\tau)/5}, \quad (5)$$

where $\text{cov}(\tau) = 1 - \text{abstain_rate}(\tau)$, and any residual tie is settled in favour of the lowest τ , on the grounds that a lower threshold answers more questions for the same decision quality. The rule lives in a standalone script so that anyone can re-run it and get the same number.

E. Deriving the Sweep Rather Than Re-Running It

Neither retrieval nor generation depends on τ . The same query retrieves the same passages and, when answered at all, produces the same answer at every threshold. So each query is run exactly once and the entire sweep is computed afterwards by masking on the cached confidence scores.

This is roughly k times cheaper in API calls, but the more important consequence is that it makes the sweep exact. Re-generating and re-judging at every threshold, which is the obvious way to write this experiment, lets nondeterminism in the judge move the faithfulness column independently of the variable under study. Some of the wobble we saw in earlier versions of this table was almost certainly that, and not a real effect of the threshold.

F. Models

Generation runs on Groq-hosted `openai/gpt-oss-20b`. Evaluation uses `openai/gpt-oss-120b` as the judge, held fixed across every experiment reported in this paper, so all scores are mutually comparable. Using a larger model to grade a smaller one also avoids the degenerate case of a generator marking its own work. Both run at temperature 0.

One practical note for anyone reproducing this. The `gpt-oss` family are reasoning models that spend completion tokens on a separate `reasoning` field before emitting anything in content. If `max_tokens` is set tightly, the budget is consumed by reasoning and `content` comes back empty, which looks like a parsing bug and is not. We use a limit of 1024.

IV. EXPERIMENTAL SETUP

A. Corpus

We scraped 62 Wikipedia pages covering the Indian Premier League: the league itself and its governing body, all ten current franchises, five defunct ones, twenty prominent players, six season pages, four records and statistics pages, eight venues, and a handful of pages on the surrounding cricket context. That produced 823 sections, which were split into 2,732 passages of roughly 500 characters with 100 characters of overlap between neighbours.

The IPL was chosen for a specific reason rather than an arbitrary one. It is factually dense, it is well covered on Wikipedia, and its boundaries are sharp enough that we could sit down and decide by hand which questions the corpus can and cannot answer. That last property is what makes ground-truth answerability labels possible at all, and without those labels none of the abstention metrics in this paper could be computed.

Two details of the scraping are worth recording because they are easy to get wrong. Sections are collected recursively rather than at the top level only: on Wikipedia a heading such as *History* or *Rivalries* usually contains no prose of its own, with everything substantive sitting one or two levels below it, so a top-level walk silently discards most of every page. And page titles are resolved to their canonical form before scraping, with a guard against redirect collisions, because several plausible IPL page titles redirect onto pages already collected and would otherwise store the same text twice.

B. Query Set

We wrote 160 questions by hand, each tagged with a category and a ground-truth answerability label (Table I). The composition is the most important design decision in this paper, so it is worth walking through.

Ninety questions are ordinary in-domain questions that the corpus should be able to answer, spread across four types: simple factual lookups, reasoning questions that ask why or how something happened, comparisons between two entities, and numeric questions. These are labelled *true*.

Eighteen questions have nothing to do with cricket at all, covering science, geography, history and the like (“*What is the capital of Mongolia?*”). They are unanswerable by construction and labelled *false*. In earlier versions of this work these were the only unanswerable questions, and that turned out to be the flaw.

Thirty-five questions are what we call *near-miss*. They sound like entirely reasonable IPL questions but ask for something Wikipedia does not carry. They fall into recognisable families: questions about the future (“*Which players will Chennai Super Kings retain for the 2027 auction?*”), hyper-specific statistics that no encyclopedia records (“*What was Bumrah’s economy rate in powerplay overs during the 2022 IPL?*”), behind-the-scenes matters (“*What is the internal scouting process used by Kolkata Knight Riders?*”), granular match details (“*Who bowled the eighteenth over of the 2023*”).

final?”), obscure conditional records (“Which bowler has the best figures in an IPL match played on a Tuesday?”), and commercially confidential information (“What is the revenue sharing formula between IPL franchises and the BCCI?”). These are labelled *false*. They are the heart of the evaluation.

Seventeen questions are *in-domain hard*: IPL questions asking for exact figures that Wikipedia sometimes carries and sometimes does not, such as a season’s total attendance or a player’s strike rate in finals. We could not honestly certify these either way, so they are labelled *uncertain* and excluded from every abstention metric. They are still generated and scored, because they tell us something about the generator’s own behaviour.

Every label was checked against the corpus before use. For the answerable questions we confirmed the supporting text exists; for the near-miss questions we read the passages the retriever actually returns and confirmed the answer is not among them. That check caught one mistake worth mentioning, because it illustrates how easily this goes wrong. We had written “What do cricket experts think about the impact player rule?” as a near-miss, on the assumption that Wikipedia does not editorialise. It does: the 2023 season page states that “many cricket pundits criticised the new impact player rule.” The question was answerable, and we replaced it.

TABLE I
QUERY SET COMPOSITION (N = 160)

Category	n	Answerable	Purpose
Factual	40	true	retrieval/generation
Reasoning	20	true	retrieval/generation
Comparative	15	true	retrieval/generation
Numeric	15	true	retrieval/generation
In-domain hard	17	uncertain	stress test
Near-miss	35	false	abstention (hard)
Out-of-domain	18	false	abstention (easy)
Labelled for abstention scoring		143	

C. Baseline

The baseline is the same generation model, handed the question with no retrieved context and no option to abstain. Holding the model fixed is what isolates grounding as the variable: whatever faithfulness gap appears between RAG and the baseline cannot be attributed to one model simply being stronger than the other.

The baseline is judged against the same retrieved context the RAG system received. This is deliberate and slightly counter-intuitive, so it deserves a sentence. The question faithfulness asks is whether an answer is supported by the passages, not whether it happens to be true in the world. Judging the baseline against those same passages is what makes the two faithfulness columns comparable, and it is precisely how an ungrounded but factually correct answer earns a low score.

D. Metrics

- **Relevance** and **faithfulness**, each scored 1–5 by the judge. An abstention is assigned relevance 1 and faith-

fulness 5 without consulting the judge, on the reasoning that it did not answer but also did not invent anything.

- **Effective relevance** and **effective faithfulness**: the same two scores averaged over answered queries only. We report these alongside the overall figures throughout, because the overall faithfulness number is inflated by the scoring rule that hands every abstention a 5, and a reader deserves to see the un-inflated version next to it.
- **Hit@ k** , **Precision@ k** and **MRR@ k** , computed by keyword matching against the 90 answerable queries that carry ground-truth keywords.
- **Abstention precision, recall, F1, specificity and accuracy**, with “should abstain” as the positive class, over the 143 queries carrying a definite label.
- **Matthews correlation coefficient**, which uses all four cells of the confusion matrix and so penalises both error types proportionally even under class imbalance:

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}. \quad (6)$$

- **Hallucination rate**: the share of substantive answers scored below 3 on faithfulness. Refusals are excluded from both the numerator and the denominator, for reasons set out in Section V-J.

V. RESULTS

A. RAG With Abstention Against the Baseline

Table II sets RAG at $\tau^* = 0.62$ against the no-context baseline, and Fig. 1 shows the same comparison.

Faithfulness more than doubles, from 2.19 to 4.44. Restricting to answered queries only, so that the abstention scoring rule contributes nothing, it is still 4.18. The hallucination rate tells the same story more bluntly: 7 unfaithful answers out of 84 substantive ones for RAG, against 99 out of 160 for the baseline.

The baseline wins on raw relevance, 4.23 against 3.08, and this is not a defect in the RAG system. The baseline answers all 160 questions, including the 53 it has no possible grounding for, and it answers them fluently from pretrained knowledge. That is exactly the behaviour faithfulness is designed to catch, and the two columns should be read together rather than separately. Among the 109 questions RAG actually answered, its relevance is 4.06.

TABLE II
RAG WITH ABSTENTION VS. NO-CONTEXT BASELINE (N = 160,
 $\tau^* = 0.62$)

Metric	RAG	Baseline
Relevance (all 160)	3.08	4.23
Relevance (answered only, $n = 109$)	4.06	–
Faithfulness (all 160)	4.44	2.19
Faithfulness (answered only)	4.18	–
Hallucinations (faithfulness < 3)	7/84	99/160
Hallucination rate	0.08	0.62
Abstain rate	0.32	0.00
Generator-side refusals	25/109	–

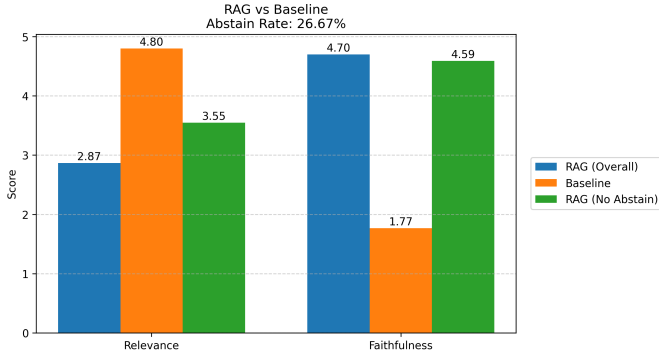


Fig. 1. RAG with abstention against the no-context baseline. The answered-only bars exclude the 51 abstentions, so they show the scores without the contribution of the rule that assigns every abstention a faithfulness of 5.

B. Retrieval Quality

Over the 90 answerable queries, Hit@3 is 0.93, Precision@3 is 0.73 and MRR@3 is 0.85. Retrieval finds a relevant passage for the large majority of answerable questions and usually ranks it first.

None of these figures vary with τ , and that is structural rather than empirical: the threshold governs whether retrieved context gets used, not what gets retrieved. The residual 0.27 in Precision@3 is the more interesting number, because it says that roughly one of every three passages the generator receives is not relevant, and that is the raw material from which the confident-but-wrong answers in Section V-G are assembled.

C. What the Confidence Score Actually Separates

This is the central result, and Fig. 2 carries it.

Plotting every query’s confidence by its ground-truth class shows two completely different pictures depending on which unanswerable class you look at. Out-of-domain questions score between 0.135 and 0.336. Answerable questions score between 0.517 and 0.824. Nothing whatsoever falls in the 0.18-wide band between them, which means any threshold placed inside that band classifies all 108 of those queries correctly. For this class the mechanism is not merely good, it is exact.

Near-miss questions score between 0.462 and 0.715, and that range sits inside the answerable range. Seventy-four of the 90 answerable questions score no higher than the best-scoring near-miss question. There is no gap to put a threshold in, and consequently no threshold that separates the two classes. Whatever value of τ is chosen, refusing more near-miss questions necessarily means refusing more answerable ones at close to the same rate.

The reason is not mysterious once stated. Retrieval similarity measures topical proximity, and a near-miss question is topically proximate by construction. Asking what Bumrah’s powerplay economy rate was in 2022 retrieves passages about Bumrah’s economy rate in other contexts, which are a very good topical match for the question and contain nothing that answers it. The confidence score has no way to see the difference, because the difference is not topical. It is about

whether a specific claim is present, and similarity does not encode that.

D. Threshold Sweep

Table III gives the full sweep and Fig. 3 plots it. The shape is a genuine coverage–reliability tradeoff, which is what the overlap in Section V-C implies it has to be.

Recall on the abstention target climbs monotonically with τ : it is 0.00 at 0.10, reaches 0.32 by 0.30, 0.72 at the selected threshold, and only saturates at 1.00 by $\tau = 0.75$. Precision runs the other way, holding at 1.00 through $\tau \leq 0.50$ and falling to 0.38 by 0.80. The two curves cross in the low 0.60s, and MCC peaks there at 0.65, which is what selects $\tau^* = 0.62$.

We want to flag an earlier mistake of our own here, because it is instructive. A previous version of this study used only out-of-domain questions as its unanswerable class, and over that query set recall was pinned at 1.00 across every threshold we tested, with only precision moving. The natural conclusion was that there is no real tradeoff in this mechanism, only precision decay above a saturation point. That conclusion was an artifact of the evaluation set. Add unanswerable questions that are hard rather than obvious, and the tradeoff appears immediately. A sweep whose headline metric never moves has not established anything about that metric, and we would encourage anyone reporting saturated abstention recall to check whether their unanswerable class is simply too easy.

Two further columns are worth a look. Effective relevance rises with τ , from 3.30 to 5.00, but this is a selection effect and not an improvement in generation: as harder questions get abstained on, the ones left answered are on average the easier ones. And the hallucination rate among substantive answers is roughly flat through the usable range, moving from 0.12 to 0.08 between $\tau = 0.10$ and τ^* , which is the basis for the claim in Section VI that the faithfulness gain comes from grounding rather than from abstention.

E. The Abstention Decision at τ^*

At the selected threshold the confusion matrix over the 143 labelled queries is TP = 38, FP = 8, TN = 82, FN = 15, giving precision 0.83, recall 0.72, F1 0.77, specificity 0.91, accuracy 0.84 and MCC 0.65.

Specificity of 0.91 says the mechanism is not trigger-happy: it preserves nine of every ten answerable questions. Recall of 0.72 says the other side of the ledger is where the trouble is, and all 15 false negatives are near-miss questions, since every out-of-domain question is caught. Read together, the two numbers describe a mechanism that is good at not getting in the way and mediocre at catching the hard cases.

F. Results by Category, and a Second Refusal Channel

Breaking the results down by category (Table IV) surfaces something the aggregates hide.

The abstain rate falls exactly where the separation analysis predicts. Out-of-domain questions are refused 100% of the time. Near-miss questions are refused 57% of the time. Answerable categories sit between 0.07 and 0.13, which is the false-positive cost of operating at τ^* .

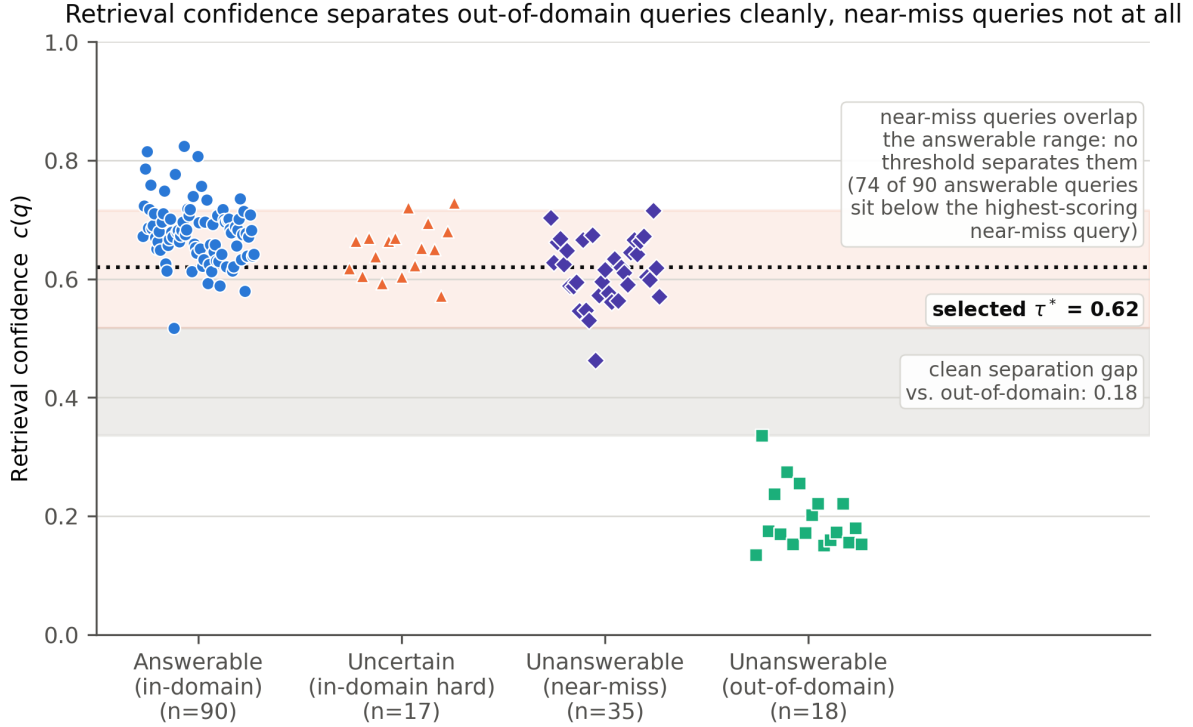


Fig. 2. Retrieval confidence by ground-truth class, one point per query. Out-of-domain queries are separated from answerable ones by a clean margin of 0.18, so any threshold inside that band classifies all 108 of them correctly. Near-miss queries overlap the answerable range entirely — 74 of the 90 answerable queries score no higher than the best-scoring near-miss — so no scalar threshold can separate those two classes at any operating point.

The column worth dwelling on is generator refusals. These are queries the gate let through, on which the generator declined to answer anyway, replying “I don’t know” rather than inventing something. There are 25 of them among the 109 answered queries, and they are not spread evenly. Twelve are near-miss and seven are in-domain hard, which is to say that 19 of 25 fall in precisely the two categories the confidence gate handles worst.

So the system has two independent refusal channels, and they fail in different places. The confidence gate detects *topic absence*: it knows when nothing in the corpus is about this subject. The generator detects *claim absence*: handed passages that are topically right but do not contain the specific fact asked for, it often notices and says so. Neither alone is sufficient, and the second one is doing quiet work that a paper reporting only abstain rates would miss entirely. It also explains why in-domain hard questions score 4.29 on faithfulness despite scoring only 2.12 on relevance: they are largely not being answered wrongly, they are not being answered at all.

G. Confident but Wrong

Seven substantive answers passed the gate and were still judged unfaithful. They are worth looking at individually, because they show what the mechanism cannot see.

Asked which franchise Red Chillies Entertainment owns, the system answered “Lucknow Super Giants” at confidence 0.663. The correct answer is Kolkata Knight Riders, and the

corpus says so. Asked how many teams played the inaugural IPL season, it answered “five” at confidence 0.675; the answer is eight. In both cases retrieval returned passages about IPL ownership and IPL seasons respectively, which are topically correct and high-scoring, and did not contain the specific fact. The gate saw a confident retrieval and let it through.

The sharpest example is a near-miss question. Asked “Which bowler has the best figures in an IPL match played on a Tuesday?”, the system replied that it was Axar Patel with 11 wickets for 70 runs against England on 13 February 2021, at confidence 0.642. That is a real bowling record, correctly transcribed from a retrieved passage. It is from a Test match, not an IPL match, and has nothing to do with Tuesdays. The retrieved text was topically plausible, the model assembled a fluent answer from it, and the confidence gate had no basis on which to object.

This is the failure the threshold cannot address, and it is a different failure from the one it does address. Catching it would need a signal about whether a specific claim is supported, not whether a topic is present, which means either verification of the generated claim against the passages or an uncertainty estimate from the generator itself.

H. Confidence Variants and a Chance Baseline

Table V compares the three confidence definitions from Section III-C and adds a chance-level comparison.

TABLE III

THRESHOLD SWEEP (N = 160). ABSTENTION METRICS ARE COMPUTED OVER THE 143 QUERIES CARRYING A DEFINITE ANSWERABILITY LABEL. FAITH. (ANS.) IS FAITHFULNESS OVER ANSWERED QUERIES ONLY; HALL. IS THE HALLUCINATION RATE AMONG SUBSTANTIVE ANSWERS.

τ	Rel.	Eff. Rel.	Faith.	Faith. (ans.)	Abstain	Hall.	P	R	F1	Spec.	Acc.	MCC
0.10	3.30	3.30	3.58	3.58	0.00	0.12	0.00	0.00	0.00	1.00	0.63	0.00
0.15	3.30	3.31	3.60	3.59	0.01	0.12	1.00	0.02	0.04	1.00	0.64	0.11
0.20	3.30	3.47	3.85	3.77	0.07	0.12	1.00	0.21	0.34	1.00	0.71	0.38
0.25	3.30	3.54	3.95	3.84	0.09	0.12	1.00	0.28	0.44	1.00	0.73	0.45
0.30	3.30	3.57	4.00	3.88	0.11	0.12	1.00	0.32	0.49	1.00	0.75	0.48
0.35	3.30	3.59	4.01	3.89	0.11	0.12	1.00	0.34	0.51	1.00	0.76	0.49
0.40	3.30	3.59	4.01	3.89	0.11	0.12	1.00	0.34	0.51	1.00	0.76	0.49
0.45	3.30	3.59	4.01	3.89	0.11	0.12	1.00	0.34	0.51	1.00	0.76	0.49
0.50	3.30	3.61	4.04	3.91	0.12	0.12	1.00	0.36	0.53	1.00	0.76	0.51
0.55	3.27	3.66	4.11	3.96	0.14	0.11	0.96	0.42	0.58	0.99	0.78	0.53
0.60	3.22	3.93	4.33	4.11	0.24	0.11	0.89	0.62	0.73	0.96	0.83	0.64
0.62*	3.08	4.06	4.44	4.18	0.32	0.08	0.83	0.72	0.77	0.91	0.84	0.65
0.65	2.76	4.36	4.62	4.27	0.47	0.07	0.65	0.83	0.73	0.73	0.77	0.54
0.70	1.66	4.53	4.95	4.73	0.81	0.04	0.44	0.96	0.61	0.29	0.54	0.31
0.75	1.18	5.00	5.00	5.00	0.96	0.00	0.39	1.00	0.56	0.08	0.42	0.17
0.80	1.07	5.00	5.00	5.00	0.98	0.00	0.38	1.00	0.55	0.03	0.39	0.11

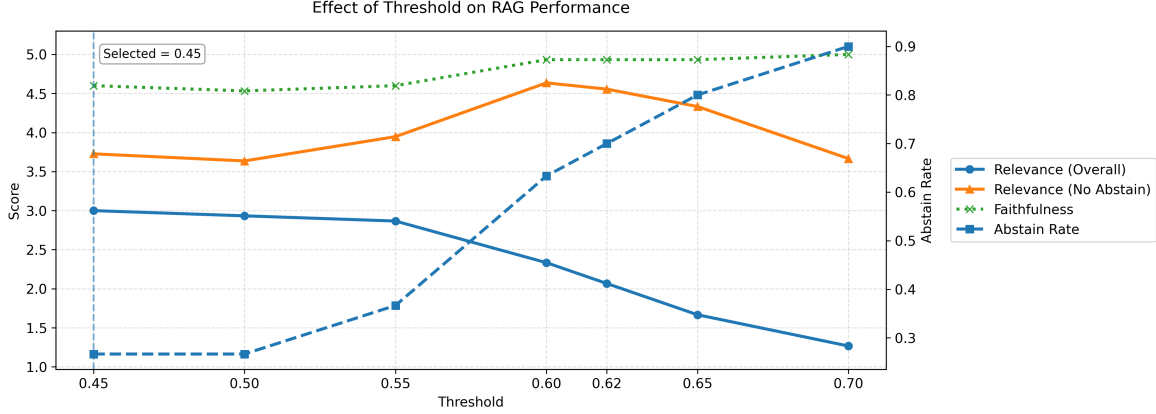


Fig. 3. Effect of the abstention threshold. Judge scores on a 1–5 scale are plotted above, rates and coefficients on a 0–1 scale below, sharing one x axis. Precision and recall cross in the low 0.60s, where MCC peaks; the dotted line marks the selected $\tau^* = 0.62$, read from the selection script rather than annotated by hand.

TABLE IV

RESULTS BY CATEGORY AT $\tau^* = 0.62$. “GEN. REF.” COUNTS QUERIES THE GATE PASSED BUT THE GENERATOR DECLINED.

Category	n	Abst.	Gen. ref.	Rel.	Faith.	Mean $c(q)$
Factual	40	0.07	1	4.62	4.67	0.691
Reasoning	20	0.10	2	4.20	4.50	0.670
Comparative	15	0.13	3	3.53	3.40	0.666
Numeric	15	0.07	0	4.73	4.73	0.669
In-domain hard	17	0.29	7	2.12	4.29	0.650
Near-miss	35	0.57	12	1.31	4.26	0.613
Out-of-domain	18	1.00	0	1.00	5.00	0.196

Neither alternative beats the plain mean. Rank-weighting gives up recall (0.66 against 0.72) for a marginal specificity gain and ends at F1 0.74. The maximum shifts the operating point rather than improving it, trading precision for recall to land at F1 0.75. The mean’s advantage is small and we would not want to over-read it, but the useful conclusion is negative and still worth reporting: the obvious refinements to

the scoring rule do not rescue the mechanism, because the problem is not how the three scores are combined. It is that all three scores are measuring topical similarity, and the near-miss queries are topically similar.

The chance baseline matters more. At τ^* the system abstains on 46 of the 143 labelled queries, so we compared against abstaining on 46 labelled queries chosen uniformly at random, averaged over 100 seeds. Random abstention scores F1 0.35 and MCC 0.00. The confidence gate scores 0.77 and 0.65. The mechanism is unambiguously doing real work, and stating that against an explicit chance level seems better practice than leaving the reader to guess what a given F1 is worth on an imbalanced set.

I. Validating the Judge

An LLM judge is a measuring instrument, and an unvalidated instrument is not worth much. One author hand-scored 15 items against the same rubric the judge uses, without looking at the judge’s scores first. The sample was drawn with a fixed seed, stratified across query categories so that

TABLE V
CONFIDENCE DEFINITIONS AND CHANCE-LEVEL ABSTENTION. THE
RANDOM ROW ABSTAINS ON THE SAME NUMBER OF LABELLED QUERIES
AS THE GATE DOES AT τ^* , AVERAGED OVER 100 SEEDS.

$c(q)$	τ^*	P	R	F1	Spec.	MCC
Mean (ours)	0.62	0.83	0.72	0.77	0.91	0.65
Rank-weighted	0.62	0.83	0.66	0.74	0.92	0.62
Maximum	0.65	0.73	0.77	0.75	0.83	0.60
Random (chance)	—	0.37	0.32	0.35	0.68	0.00

the harder ones were represented, and generator refusals were excluded because faithfulness is undefined for an answer that asserts nothing.

Results are in Table VI. Agreement is substantial on both axes: Pearson r of 0.83 for relevance and 0.85 for faithfulness, Spearman ρ of 0.76 and 0.66, and quadratic-weighted κ of 0.80 and 0.83. Mean absolute error is 0.40 on a five-point scale for both, and 22 of the 30 individual judgements match the human rating exactly.

We report κ and ρ alongside r deliberately. Ratings on this kind of data pile up at the ceiling, which inflates exact-match rates and leaves Pearson r to be determined by a handful of disagreements. Weighted κ corrects for chance agreement and penalises larger gaps more heavily, and it is the number we would trust most here.

The eight disagreements are informative about where the judge is soft. Three of them are faithfulness cases where the human rater judged an answer better supported than the judge did, suggesting the judge is somewhat strict about grounding. Three more involve in-domain hard questions, where deciding how much credit a partial answer deserves is genuinely difficult and two careful raters could reasonably differ. None of the disagreements is larger than two points.

We note that $n = 15$ with one rater is a small validation, and we would not want it carrying more weight than it can bear. It is enough to establish that the judge is not measuring something unrelated to human judgement, which is the claim we make from it, and not enough to characterise its behaviour precisely.

TABLE VI
HUMAN VS. LLM-JUDGE AGREEMENT ($n = 15$ ITEMS, 30 JUDGEMENTS)

Metric	MAE	Exact	Pearson r	Spearman ρ	κ_w
Relevance	0.40	0.73	0.83	0.76	0.80
Faithfulness	0.40	0.73	0.85	0.66	0.83

J. A Refusal Is Not a Hallucination

One finding came out of the experiments unasked for, and it changes how the headline reliability metric should be computed.

Across the run the generator produced 66 refusals. The judge scored these inconsistently: 42 received faithfulness 1, twenty-three received 5, and one received 3. The identical

string "I don't know." drew all three scores on different queries, under one judge model at temperature 0.

This is not really the judge misbehaving. The faithfulness rubric asks whether an assertion is supported by the context, and a refusal makes no assertion, so the rubric has no defined behaviour on one. The judge is being asked an ill-posed question and is answering it differently each time.

The practical consequence is concrete. If refusals are left in the hallucination count, twelve of them fall below the cutoff and the RAG hallucination count goes from 7 to 19, an overstatement of roughly 170% in the paper's central reliability metric, driven entirely by a rubric gap rather than by anything the system did. We therefore exclude refusals from both the numerator and the denominator of every hallucination rate, applying the same rule to RAG and baseline alike so the comparison stays fair.

There is a secondary point here that costs nothing to make. This is direct, reproducible evidence about the reliability of LLM-as-judge evaluation, obtained without any human annotation: the judge is self-inconsistent on a well-defined and easily detected class of input. Anyone using an LLM judge on a task where refusals or non-answers occur should check for this rather than assume it away.

VI. DISCUSSION

Three things follow from these results that we think generalise beyond our corpus.

The difficulty of your unanswerable class determines what you can conclude. This is the finding we would most want a reader to take away. With only off-topic questions, retrieval confidence separates perfectly, every abstention metric saturates, and the mechanism looks close to solved. Add questions that are on-topic but unanswerable and recall drops to 0.72 with no threshold available to fix it. Both results came from the same code, the same corpus and the same model. Only the evaluation set changed. Reported abstention performance is therefore a claim about the query set at least as much as about the method, and papers in this area should be read with that in mind, ours included.

Similarity-based confidence detects topic absence, not answer absence. These are different properties and the score only carries the first. This sets a ceiling that no amount of tuning the threshold, weighting the ranks, or swapping mean for max will lift, which is what Table V shows empirically. Getting past it requires a qualitatively different signal: verification of the generated claim against the retrieved passages, or an uncertainty estimate from the generator. The evidence in Section V-F that the generator already refuses 19 of the hard cases on its own suggests that second signal exists and is worth extracting deliberately rather than relying on as a side effect.

The faithfulness gain comes from grounding, not from abstention. This distinction is easy to lose. Faithfulness rises from 2.19 to 4.44, and it is tempting to credit the abstention mechanism. But the hallucination rate among substantive answers is roughly flat across the usable range of τ , moving

only from 0.12 to 0.08. What abstention removes is a set of queries that would have been answered ungroundedly; it does not improve the answers that remain, and Section V-G shows it cannot catch a confident retrieval failure on a question that is in-domain. The honest formulation is that grounding reduces hallucination and abstention reduces exposure, and a system designer should know which of the two they are buying.

For practitioners choosing a threshold, the concrete advice is this: measure the confidence distributions of your own answerable and unanswerable queries before choosing τ , and look specifically at whether they overlap. If they separate cleanly, a wide range of thresholds works and the choice is not delicate. If they overlap, there is no safe threshold, the choice is a real tradeoff between coverage and reliability, and it should be made against labelled answerability data rather than against relevance and faithfulness curves, which vary smoothly and hide the structure underneath.

VII. LIMITATIONS

- **One domain, one corpus.** Everything here rests on 62 IPL-related Wikipedia pages. Whether the pattern holds on a larger, noisier or less factually dense corpus is untested. Encyclopedic prose is unusually clean, and we would expect the separation between out-of-domain and in-domain to be less crisp on messier text.
- **One run, no confidence intervals.** With $n = 160$ and a single run, a single query is 0.6% of the set and 2.9% of the near-miss class. We report point estimates and no error bars, and small differences between adjacent thresholds should not be over-read.
- **The near-miss class is our construction.** We wrote these questions to be unanswerable, and verified them against the corpus, but they are still our idea of what a hard unanswerable question looks like. Naturally occurring unanswerable queries from real users may be distributed differently, and would be a better test.
- **The confidence score is uncalibrated.** It is a raw cosine average, not a probability. The thresholds reported here are specific to `all-mpnet-base-v2` on this corpus and will not transfer numerically to another setup.
- **The faithfulness rubric is partly circular for abstentions.** An abstention is assigned faithfulness 5 by definition, so some of the improvement at higher abstain rates reflects that rule rather than an independently observed effect. This is why we report effective faithfulness alongside the overall figure throughout.
- **The judge is undefined on refusals.** Documented in Section V-J. We handle it by exclusion, which is defensible but is a decision, not a measurement.
- **The human validation is small.** Fifteen items, one rater, no inter-annotator agreement estimate. It supports the claim that the judge tracks human judgement and does not support finer claims than that.
- **Keyword-based retrieval metrics are approximate.** $\text{Hit}@k$ and $\text{Precision}@k$ are computed by substring matching against hand-written keywords, which will oc-

asionally miss a passage that answers the question in different words. We did not tune keywords to improve these numbers.

- **No reranking or hybrid retrieval.** Retrieval is a single dense pass. A cross-encoder reranker [8] or a BM25 hybrid [7] would likely raise $\text{Precision}@3$ above 0.73, and since that residual is where the confident-but-wrong answers originate, it might reduce them. That is untested here.

VIII. CONCLUSION AND FUTURE WORK

We studied a confidence-gated abstention mechanism for RAG on a 160-question evaluation set built deliberately to contain unanswerable questions of two different difficulties. Against a no-context baseline, faithfulness rises from 2.19 to 4.44 out of 5 and the hallucination rate falls from 0.62 to 0.08. Scored as a classification problem against ground truth, the abstention decision reaches F1 0.77 and MCC 0.65 at the selected threshold, well clear of a chance-matched baseline at 0.35 and 0.00.

The finding we consider most useful is about the boundary of the method rather than its performance. Retrieval confidence separates off-topic questions from answerable ones with a clean margin and catches all of them. It does not separate near-miss questions at all: their confidence range sits inside the answerable range, 74 of 90 answerable questions score no higher than the best near-miss, and no threshold exists that would fix this. A threshold on retrieval similarity can tell you that a topic is missing from your corpus. It cannot tell you that an answer is missing from a topic that is present.

Several directions follow. The most direct is to extract the generator’s own refusal behaviour as a deliberate second signal rather than leaving it as a side effect, since Section V-F shows it already catches 19 of the cases the gate misses. Claim-level verification against the retrieved passages is the more thorough version of the same idea. Beyond that: repeating the study on a corpus whose domain boundary is genuinely blurry, where we expect even the out-of-domain separation to degrade; adding reranking to attack the 0.27 of retrieved passages that are irrelevant; calibrating the confidence score so that thresholds transfer between corpora; and running a larger, multiply-annotated judge validation than the 15 items we managed here.

REPRODUCIBILITY

All code, the query set with its labels, the scraped corpus, the per-query results and the scripts that regenerate every table and figure in this paper are available in the project repository. The threshold is selected by a standalone script rather than chosen by hand, the figures read the selected value from that script’s output so they cannot drift from it, and every number reported above regenerates from a single cached results file. Generation used `openai/gpt-oss-20b` and evaluation `openai/gpt-oss-120b`, both at temperature 0, with `all-mpnet-base-v2` embeddings and $k = 3$.

One reproducibility detail is worth passing on, since it cost us time. An earlier version of the retriever deduplicated passages through a Python `set`, which destroys rank order and, because string hashing is randomised per interpreter run, produced different orderings between processes. $\text{MRR}@k$ is not computable at all until that is fixed, and any rank-sensitive metric computed over a set rather than a list should be treated with suspicion.

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